

Alexander DeLise

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Education

Florida State University

Tallahassee, FL

B.S. Computational Science with Honors, GPA: 4.00/4.00

2023–2027

- Thesis: Active Learning with Conditional Generative Compressed Sensing
- Advised by Dr. Nick Dexter through the Honors in the Major Program
- Selected Coursework: Computational Probabilistic Modeling, High Performance Computing, Monte Carlo Methods, Data Science Meets Health Science

Florida State University

Tallahassee, FL

B.S. Applied and Computational Mathematics, GPA: 4.00/4.00

2023–2027

- Activities: Undergraduate Research Opportunity Program, Presidential Scholars Program, *President*, Pi Mu Epsilon Mathematics Honor Society, Society for Industrial and Applied Mathematics
- Selected Coursework: Advanced Calculus I and II (Analysis)

Academic Experience

Florida State University

Tallahassee, FL

Undergraduate Researcher - Generative Compressed Sensing

November 2026 – Present

- Advised by Dr. Nick Dexter
- Research topics: compressed sensing, generative modeling, inverse problems, active learning, scientific machine learning, high-dimensional representation learning
- Languages and tools: Python, PyTorch

Emory University

Atlanta, GA

Research Fellow – Scientific ML and Data Science

June 2025 – August 2025

- Advised by Dr. Matthias Chung
- Research topics: scientific machine learning, inverse problems, low-rank matrix approximation, Bayes risk minimization, rank-constrained regression
- Languages and tools: Python, PyTorch

University of Tennessee, Knoxville

Knoxville, TN

Research Fellow – Quantum Algorithms

May 2024 – April 2025

- Advised by Dr. James Ostrowski
- Research topics: quantum computing, variational quantum algorithms, constrained optimization
- Languages and tools: Python, PennyLane, PyTorch, Pandas

FAMU-FSU College of Engineering

Tallahassee, FL

Undergraduate Researcher – Operations Research

September 2023 – June 2025

- Advised by Dr. Arda Vanli
- Research topics: operations research, industrial engineering, mixed-integer nonlinear programming, data-driven healthcare, disease modeling
- Languages and tools: Python, Gurobi, GAMS, IBM ILOG CPLEX

Industry Experience

Johns Hopkins Applied Physics Laboratory

Data Science Intern

Laurel, MD

Incoming May 2026

– Incoming Data Science intern in the KBS Systems Performance Analysis group

Publications

Conference Publications

1. **A. DeLise**, S. Abazari, A. Vanli. “An Epidemiological Mixed-Integer Nonlinear Programming Framework for Vaccine Modeling and Patient Allocation During Pandemics”, *Proceedings of the 10th North American International Conference on Industrial Engineering and Operations Management, Orlando, Florida, 2025*. DOI: [10.46254/NA10.20250088](https://doi.org/10.46254/NA10.20250088).

Preprints

1. **A. DeLise**, K. Loh, K. Patel, M. Teague, A. Arnold, M. Chung. “Optimal Linear Baseline Models for Scientific Machine Learning” (under review), 2025. [arXiv:2508.05831](https://arxiv.org/abs/2508.05831).
2. A. Wilkie, **A. DeLise**, A. Del Real, R. Herrman, J. Ostrowski. “Learning Feasible Quantum States for Quadratic Constrained Binary Optimization Problems” (under review), 2025. [arXiv:2508.02590](https://arxiv.org/abs/2508.02590).

Presentations

Invited Talks

1. Nov 2025. *SIAM Student Chapter*, Florida State University, Tallahassee, FL.
Optimal Linear Baseline Models for Scientific Machine Learning [\[pdf\]](#)

Contributed Talks

1. Jan 2026. *Joint Mathematics Meetings, AMS Contributed Paper Session on Computer Science, Information, and Communication*, Washington, DC.
Optimal Rank-Constrained Mappings for Linear ED Architectures

Poster Presentations

1. Jan 2026. *Joint Mathematics Meetings AMS-PME Student Poster Session*, Emory University, Atlanta, GA.
Optimal Rank-Constrained Mappings for Linear ED Architectures [\[pdf\]](#)
2. Aug 2025. *CMDs REU Poster Presentation*, Emory University, Atlanta, GA.
Optimal Rank-Constrained Mappings for Linear ED Architectures [\[pdf\]](#)
3. Apr 2025. *25th Annual Undergraduate Research Symposium*, Florida State University, Tallahassee, FL.
Data-Driven Patient Allocation Optimization with Epidemic and Vaccine Modeling [\[pdf\]](#)
4. Feb 2025. *Florida Undergraduate Research Conference*, University of South Florida, Tampa, FL.
Data-Driven Patient Allocation Optimization with Epidemic and Vaccine Modeling [\[pdf\]](#)
5. Jan 2025. *Joint Mathematics Meetings AMS-PME Student Poster Session*, Seattle, WA.
Learning Feasible States with ma-QAOA and QAOA for Constrained Optimization [\[pdf\]](#)
6. Jul 2024. *UTK Summer Research Symposium*, University of Tennessee, Knoxville, TN.
Learning Feasible States with ma-QAOA and QAOA for Constrained Optimization [\[pdf\]](#)

7. Apr 2024. *24th Annual Undergraduate Research Symposium*, Florida State University, Tallahassee, FL.
Cost-Effective Location Allocation for COVID-19 Patient Assignment in Florida: A Data-Driven Approach [\[pdf\]](#)

Honors and Awards

• Barry M. Goldwater Scholarship Institutional Nominee, Florida State University	2025
• Barry M. Goldwater Scholarship Institutional Nominee, Florida State University	2024
• Presidential Scholar, Florida State University	2023
• Pi Mu Epsilon Mathematics Honor Society, Florida State University	2023
• Bright Futures Academic Scholar, State of Florida	2023